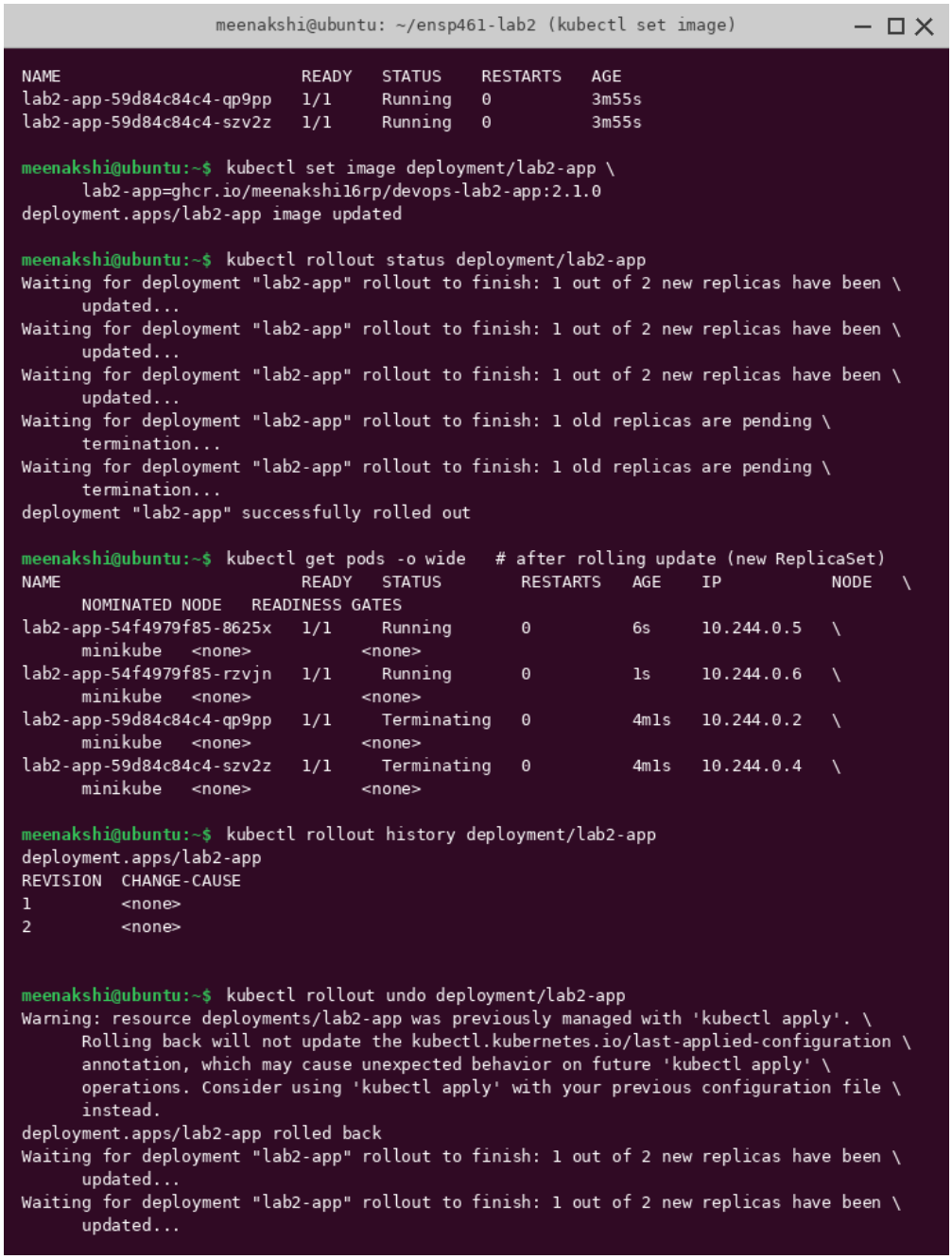
Question 6: Implement Kubernetes deployment strategies such as rolling updates and rollbacks, and demonstrate scaling of the application replicas.

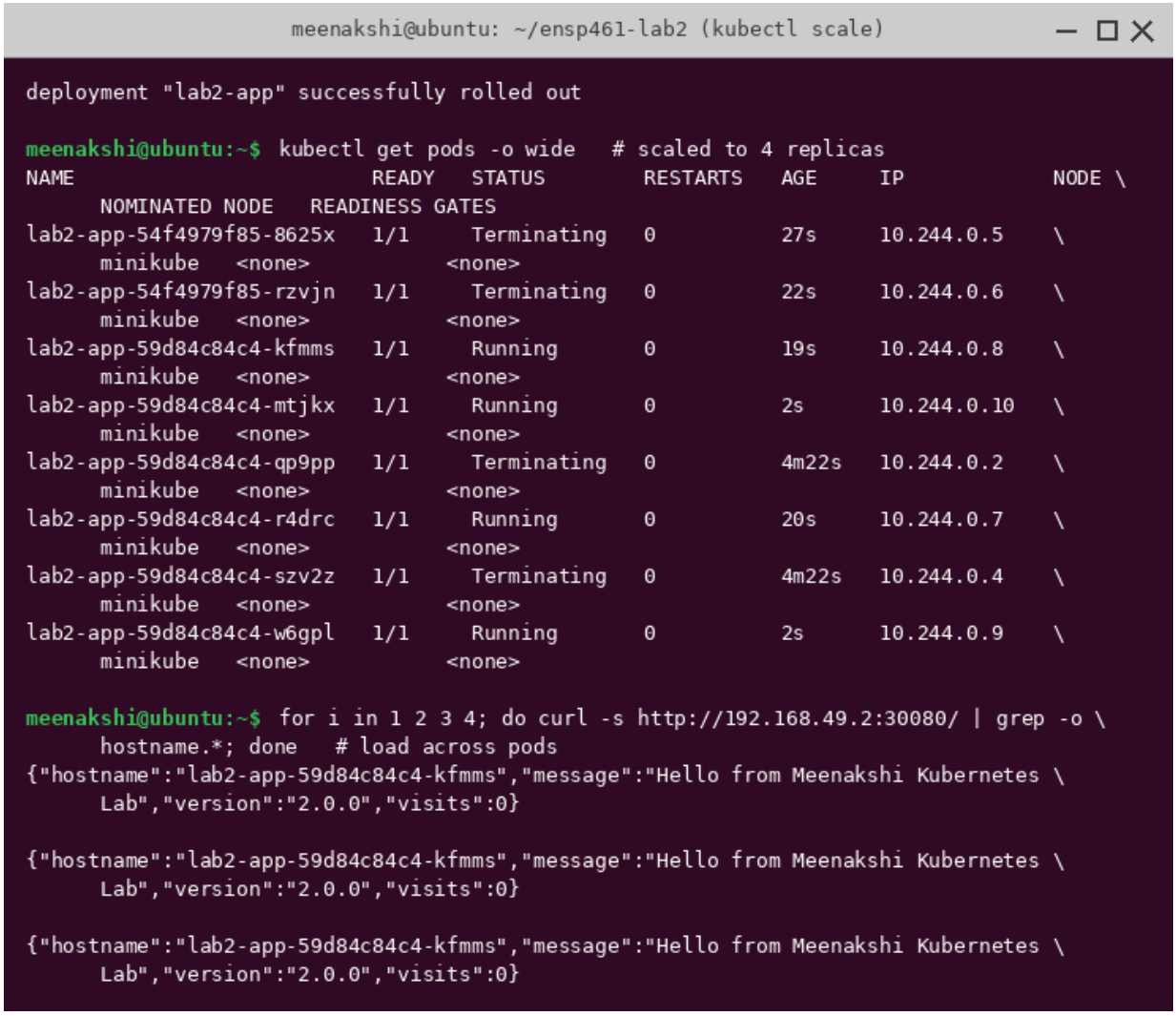
Executed on Ubuntu 24.04 LTS (WSL2) by Meenakshi Rakesh. Repository: https://github.com/meenakshi16rp/Devops-lab-2

1. Rolling update to image 2.1.0



- kubectl set image triggered a new ReplicaSet: new pods started while old pods kept serving, then terminated (visible as Running + Terminating in the same listing) - zero-downtime rolling update.
- kubectl rollout history tracks revisions 1 and 2; kubectl rollout undo returned the Deployment to image 2.0.0 (revision 3) and / served version 2.0.0 again.

2. Scaling the replicas



3. Observations

- Scaling 2 -> 4 created two additional pods and rollout status reported successfully rolled out; scaling back to 2 terminated the extra pods gracefully.
- The surge pattern (new pod Ready before old pod terminates) is controlled by the default maxSurge/maxUnavailable of the Deployment strategy.
- rollout undo does not require the old image to be re-pulled if it is still cached on the node.